3DM 2024 Proof Workshop 3: Some additional questions

Use double counting arguments for the following:

1. If A and B are finite sets, prove that

$$|A \cup B| + |A \cap B| = |A| + |B|.$$

2. If k, n are integers with $1 \leq k \leq n-1$, prove that

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

3. If k, n are integers with $0 \leq k \leq n$, prove that

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Hint: Consider the number of ways in which a committee of k people with one person being the chair can be selected from a group of n people.

4. If k, n are integers with $0 \leq m \leq k \leq n$, prove that

$$\binom{n}{k}\binom{k}{m} = \binom{n}{m}\binom{n-m}{k-m}.$$

NB: This generalizes the previous question!

5. Prove that for every nonnegative integer n,

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

6. Prove that for every nonnegative integer n,

$$\sum_{k=d}^{n} \binom{n}{k}\binom{k}{d} = 2^{n-d}\binom{n}{d}.$$

NB: This generalizes the previous question!

7. Prove that for every positive integer n,

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}.$$